

Exercise 1.

An inventory study determines that, on average, demands for a particular item at a warehouse are made 5 times per day. What is the probability that on a given day this item is requested (a) more than 5 times? (b) not at all?

Exercise 2.

For a certain manufacturing process, it is known that, on the average, 1 in every 100 items is defective. What is the probability that the fifth item inspected is the first defective item found?

Exercise 3.

An oil drilling company ventures into various locations, and its success or failure is independent from one location to another. Suppose the probability of a success at any specific location is 0.25.

- Ⓐ What is the probability that the driller drills at 10 locations and has 1 success?
- Ⓑ The driller will go bankrupt if it drills 10 times before the first success occurs. What are the driller's prospects for bankruptcy?

Exercise 4.

In an NBA (National Basketball Association) championship series, the team that wins four games out of seven is the winner. Suppose that teams A and B face each other in the championship games and that team A has probability 0.55 of winning a game over team B.

- Ⓐ What is the probability that team A will win the series in 6 games?
- Ⓑ What is the probability that team A will win the series?
- Ⓒ If teams A and B were facing each other in a regional playoff series, which is decided by winning three out of five games, what is the probability that team A would win the series?

Exercise 5.

Suppose you get on average 12 phone calls per day. Assuming phone calls are equally likely to occur at any time of day, find the probability of getting 3 phone calls in one hour.

Exercise 6.

Random variable X follows a Poisson distribution. Given that $p(0) = P(X = 0) = 1/4$, find $Var(2X - 3)$.